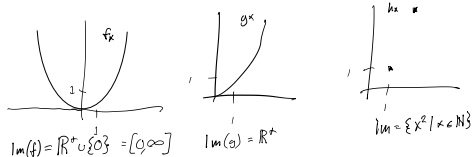


$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 \quad g: \mathbb{R}^+ \rightarrow \mathbb{R}, g(x) = x^2 \quad h: \mathbb{N} \rightarrow \mathbb{N}, h(x) = x^2$$



Injective/one-to-one functions

$$\forall x_1, x_2 \in A: (x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2))$$

Surjective/onto functions

$$\forall y \in B: \exists x \in A: f(x) = y$$

Bijjective/one-to-one correspondence

$$\text{for All } f: A \rightarrow B \\ \text{Im}(f) \subseteq B$$

$$g(x): \mathbb{R} \rightarrow \mathbb{R}$$

$$g(x) = 2x + 5$$

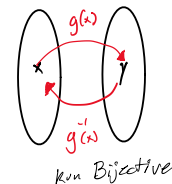
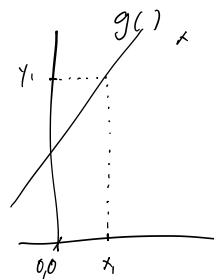
$$g^{-1}(y) = \frac{y-5}{2}$$

$$g^{-1}(g(x)) = \frac{2x+5-5}{2} = x$$

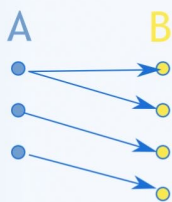
$$(f+g)(x) = f(x) + g(x)$$

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

$$(g \circ f)(x) = g(f(x))$$

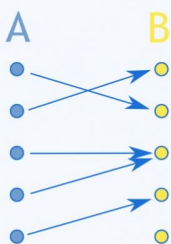


kein Bijective



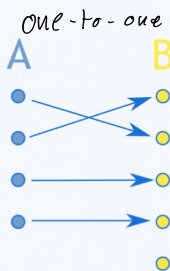
NOT a Function

A has many B



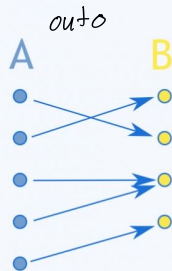
General Function

B can have many A



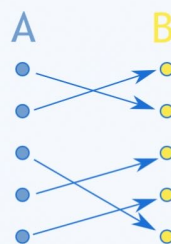
Injective (not surjective)

B can't have many A



Surjective (not injective)

Every B has some A



Bijjective (injective, surjective)

A to B, perfectly

$$(\forall y \in \mathbb{Z}: \exists u \in \mathbb{Z}: f(u) = y)$$

$$f \cdot g = (x+1) \cdot 2x$$

$$2x^2 + 2x$$

$$a_0 = 2a_{-3} - a_{-2} + a_{-1}$$

$$1 = \begin{cases} 0 \\ 2 \end{cases}$$